



Beside AP Secretariat Amaravati,
Andhra Pradesh
522237

Contretemps Nullification

Under The Guidance of:

Dr. VANACHARLA PUJITHA

Department of Mathematics

S.NO	NAME	Registration Number
1.	Pulagam Maheshwara Reddy (SCOPE)	23BCE7268
2.	Karumburu Nithin (SCOPE)	23BCE20147
3.	Ojili Subodh (SCOPE)	23BCE20245
4.	Koppali Ranjith (SCOPE)	23BCE20265
5.	Thupakula Revanth Reddy (SCOPE)	23BCE20329
6.	Shaik Abdul Raheem (SMEC)	23BME7083

Table of Contents:

Introduction.....	3
Background.....	4
Problem Identification.....	5
Objectives.....	5
Methodology.....	6
Result.....	7
Code.....	8
Conclusion.....	9
References.....	9

Introduction:

Road safety remains a significant concern, especially in regions with challenging terrains, curving roads, and limited visibility. Many accidents occur due to blind spots, sharp turns, and lane changes in restricted areas, posing a high risk for drivers and pedestrians alike. To address these challenges, the Contretemps Nullification: A Driver Warning System has been developed as an advanced yet cost-effective solution aimed at reducing road accidents and enhancing driver awareness.

This system is designed to provide real-time alerts to drivers when potential collisions or obstacles are detected. By integrating ESP-32, ultrasonic sensors, active buzzers, and LEDs, the system effectively monitors the surroundings and warns the driver when an obstruction is detected within a predefined range. The alert mechanism consists of audible signals and visual indications, ensuring quick response times to avoid possible accidents.

One of the key advantages of this system is its affordability and accessibility. With an estimated budget of ₹2500, the system offers an economical yet efficient driver assistance tool, making it viable for widespread adoption in both urban and rural areas. The modular nature of the system allows for easy integration into existing vehicles, ensuring broad usability across different driving environments.

Furthermore, the Contretemps Nullification system not only improves road safety but also enhances driver confidence, especially in conditions where natural visibility is hindered. By actively reducing collision risks, this system contributes to the overall goal of safer and smarter transportation, offering a technological step forward in accident prevention.

Through detailed circuit design, well-structured coding, and practical application, this project showcases an innovative approach to making roads safer for everyone. It stands as a testament to how technology can be leveraged to mitigate risks and promote responsible driving habits, ultimately contributing to a safer future on the roads.

Background:

With the increasing number of road accidents, especially in **hilly terrains and urban traffic**, there is a growing demand for **advanced driver assistance systems** to enhance road safety. Traditional solutions like **mirrors and cameras** often fail in **low-visibility conditions** such as fog, sharp curves, or night driving, leaving drivers at risk.

To overcome these challenges, **sensor-based warning systems** have gained popularity due to their ability to **proactively detect obstacles** and alert drivers in real time. The **Contretemps Nullification project** leverages **ultrasonic sensing technology** to detect objects in a vehicle's path and provide **immediate audio-visual warnings**, ensuring a quick response to potential hazards.

Designed as a **cost-effective and widely applicable** solution, this system utilizes **ESP-32 microcontrollers, ultrasonic sensors, buzzers, and LEDs** to enhance driver awareness. With an estimated budget of **₹2500**, it is an **affordable, efficient, and easy-to-integrate safety system**, making it suitable for both **urban and rural road conditions**. By improving **situational awareness and reducing accident risks**, this project aims to make driving **safer and more reliable** for everyone.

Problem Identification:

The project addresses the following key problems:

1. **Blind Spots & Limited Visibility** – Drivers struggle with **hidden obstacles on curved roads**.
2. **Increased Collision Risk** – Sudden lane changes in restricted areas often lead to **accidents**.
3. **Lack of Affordable Warning Systems** – **Existing systems** are expensive, making them **inaccessible to many drivers**.

Methodology:

- **Enhance Safety** – Reduce **collision risks** by providing **timely alerts** to drivers.
- **Increase Awareness** – Improve a driver's **response to obstacles** in blind spots and restricted lanes.
- **Cost-Effective Solution** – Develop an **affordable and efficient** detection system.
- **Wide Usability** – Ensure functionality in **both urban and rural environments**.
- **Accident Reduction** – Minimize the number of **road accidents** caused by **limited visibility**.

1. System Components & Design

- **ESP-32 Microcontroller** – The brain of the system, processes sensor data.
- **Ultrasonic Sensors** – Detect obstacles and measure distances.
- **Active Buzzer & LED** – Alert the driver in real-time.
- **3.7V Battery & Power Booster** – Ensure uninterrupted operation.

2. Working Mechanism

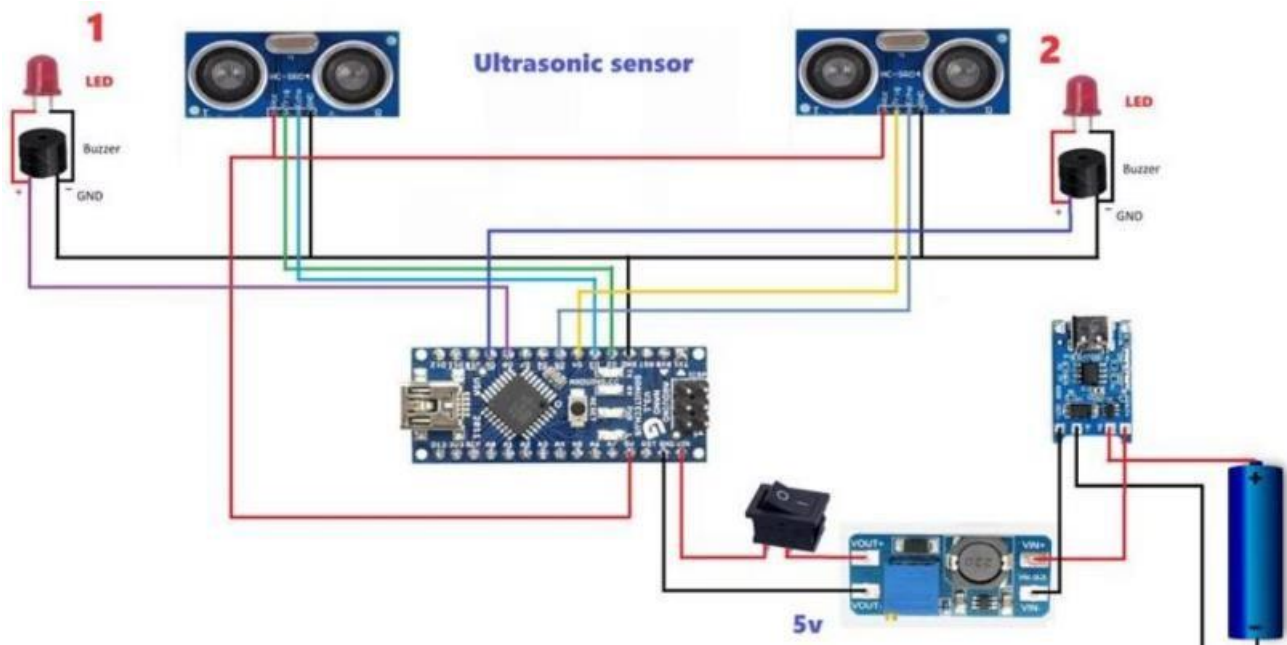
- The **ultrasonic sensors** continuously scan the surroundings.
- If an object is detected within a **predefined safety range**, the **ESP-32 processes the data**.
- The **buzzer and LED** alert the driver about the obstacle.
- The system operates efficiently even in **low visibility conditions**, ensuring **enhanced driver awareness**.

3. Implementation & Testing

- The system was **assembled and tested** in a controlled environment.
- **Obstacle detection accuracy** was evaluated by placing objects at varying distances.
- The system successfully provided **instant alerts**, confirming its **efficiency and reliability**.

Result:

- The system successfully **detected obstacles within a specified range**.
- **Real-time alerts were triggered** whenever an object was within a **dangerous proximity**.
- **Testing showed high accuracy** in distance measurement and **quick response times**. The system proved to be **cost-effective and functional**, making it a **viable safety solution** for vehicles.



Code:

1. Pin Definitions

```
const int trigPin1 = 18;  
const int echoPin1 = 21;  
const int trigPin2 = 19;  
const int echoPin2 = 22;  
const int buzzer1 = 13;  
const int buzzer2 = 14;
```

Ultrasonic Sensor 1: trigPin1 and echoPin1

Ultrasonic Sensor 2: trigPin2 and echoPin2

Buzzers: buzzer1 and buzzer2

2. Distance Threshold

```
const int distanceThreshold = 4;
```

The threshold is set to 4 cm. If an obstacle is closer than this, the buzzer will sound.

3. Setup Function

```
void setup() {  
  Serial.begin(115200);  
  pinMode(trigPin1, OUTPUT);  
  pinMode(echoPin1, INPUT);  
  pinMode(trigPin2, OUTPUT);
```

```
pinMode(echoPin2, INPUT);  
pinMode(buzzer1, OUTPUT);  
pinMode(buzzer2, OUTPUT);  
}
```

Initializes serial communication.

Configures sensor pins as input/output.

Configures buzzer pins as output.

4. Main Loop

```
void loop() {  
  long distance1 = measureDistance(trigPin1, echoPin1);  
  long distance2 = measureDistance(trigPin2, echoPin2);  
  Serial.print("Distance 1: ");  
  Serial.print(distance1);  
  Serial.print(" cm, Distance 2: ");  
  Serial.print(distance2);  
  Serial.println(" cm");  digitalWrite(mot1,1);  
  if (distance1 <= distanceThreshold) {  
    digitalWrite(buzzer1, HIGH);  
  } else {  
    digitalWrite(buzzer1, LOW);  
  }  
  if (distance2 <= distanceThreshold) {  
    digitalWrite(buzzer2, HIGH);  
  }  
}
```

```
else {  
digitalWrite(buzzer2, LOW);  
}  
delay(100);  
}
```

Measures distances from both sensors.

Displays the distances in the serial monitor.

Activates buzzers if an obstacle is detected within the threshold.

5. Distance Measurement Function

```
long measureDistance(int trigPin, int echoPin) {  
digitalWrite(trigPin, LOW);  
delayMicroseconds(2);  
digitalWrite(trigPin, HIGH);  
delayMicroseconds(10);  
digitalWrite(trigPin, LOW);  
long duration = pulseIn(echoPin, HIGH);  
long distance = duration * 0.034 / 2;  
return distance;  
}
```

Sends an ultrasonic pulse.

Reads the time taken for the pulse to return.

Converts time to distance using the formula:

$\text{distance} = \text{duration} \times 0.034 / 2$

Conclusion:

The **Contretemps Nullification** driver warning system is a **reliable, affordable, and effective** solution for improving **road safety**. By providing **real-time alerts** to drivers, the system significantly reduces the risks associated with **blind spots and limited visibility areas**. The project demonstrates **successful integration of ultrasonic sensors, microcontrollers, and alert mechanisms**, proving its potential for **real-world implementation**. Further **enhancements** could include **wireless connectivity for advanced data processing** and **integration with vehicle dashboards**.

References:AS

1. Arduino. (n.d.). Arduino - Home. Retrieved from <https://www.arduino.cc/>
2. DHT Sensor Library for Arduino. (n.d.). GitHub. Retrieved from <https://github.com/adafruit/DHTsensor-library>
3. IRFZ44N Datasheet. (n.d.). Vishay. Retrieved from <https://www.vishay.com/docs/91017/91017.pdf>
4. LiquidCrystal Library for Arduino. (n.d.). Arduino. Retrieved from <https://www.arduino.cc/en/Reference/LiquidCrystal>
5. "Temperature Controlled DC Fan using Arduino" by Circuit Digest. (n.d.). Circuit Digest. Retrieved from <https://circuitdigest.com/microcontroller-projects/temperature-controlled-dc-fan-usingarduino>
6. DHT11 Sensor Datasheet. (n.d.). Adafruit Industries. Retrieved from <https://www.adafruit.com/datasheets/Digital%20humidity%20and%20temperature%20sensor%20AM2302.pdf>

THANK
YOU